

PREFACE TO THE HANDBOOK

Purpose

Our objective in editing the *Handbook of Computational Economics* was to provide an introduction and selective overview of the rapidly emerging field of computational economics. Actually, computational economics is not currently recognized as a separate *field* in most economics departments, and only a handful of departments offer regularly listed courses in this area. However computer technology is rapidly changing the way many economists approach their research, and a large body of published work has already demonstrated the usefulness and importance of computational methods in economics. In a recent report to the National Science Foundation [Kendrick (1993)], a panel of economists surveyed the wide range of contributions that computer models have made to our discipline in areas such as economic growth, econometrics, international trade, general equilibrium theory, environmental change, game theory and mechanism design, static and dynamic optimization problems, and macroeconomics. The report also identified “many areas where computational economics offers substantial opportunities to extend the frontiers of knowledge” [Kendrick (1993, p. 269)]. The rapid growth in this area is reflected in the large number of computationally oriented articles in regular economics journals and the recent emergence of specialized journals in this area, including *Computational Economics* and Series A of the *Journal of Economic Dynamics and Control*. In 1994 the Society for Computational Economics (SCE) was founded, and starting in 1995 there will be annual SCE-sponsored international conferences in computational economics in addition to the frequent workshops and conferences that already take place on an *ad hoc* basis (e.g. NSF-supported conferences and workshops on computational methods such as the workshop hosted by Daniel McFadden and Paul Ruud at the University of California at Berkeley in August, 1994). The first international conference of the SCE was held at the IC² Institute in Austin Texas in May 1995 and the second international conference will be held at the University of Geneva in June, 1996. There is also a World Wide Web home page for the SCE at <http://www.unige.ch/ce> which serves as a reference point for other conferences, workshops, and research opportunities in computational economics.

Even though computational economics is still in its adolescence – rapidly growing and evolving – there are a number of subareas that are already quite mature and highly

developed. Therefore we felt that this is a good time to publish a handbook in this area. We view computational economics as providing an important set of tools that an increasing number of economists will need to acquire in order to understand and do state of the art research in virtually all areas of economics. In the near future it will be as difficult to do economics without having a good understanding of computing hardware and software and numerical methods as it is to do economics without a solid understanding of real analysis, probability, and statistics.

In the process of selecting topics for this *Handbook* we decided against trying to be comprehensive or encyclopedic in our coverage of the subject. Our objective was to have the *Handbook* sample a number of different styles and approaches, reflecting the breadth of computational economics as it is practiced today. In particular, articles in this volume range from very applied, policy oriented applications of computational methods to highly theoretical and mathematically complex analyses of algorithms and numerical methods. Although the main impetus for computational methods first arose in econometric applications in the last 1940's and early 1950's, we made a conscious decision not to attempt to cover econometric applications of computational methods since we regard this as the territory for the *Handbook of Econometrics*. In addition, we made a conscious decision to minimize overlap with topics that have been treated in recent handbooks in operations research and computer sciences since there are already a number of high quality handbooks in these areas [see e.g. van Leeuwen (1990) and Heyman and Sobel (1990)]. Our goal in this *Handbook* was to emphasize the unique contributions of computational methods in economics and focus on problems for which well developed solutions are not already available from the literatures in operations research, numerical methods, and computer science. Nevertheless, the strong influence of these literatures on the development of computational economics will be evident throughout the volume.

Finally, while a number of chapters in this *Handbook* cover relatively mature areas in computational economics, we also consciously included a number of chapters on more speculative "frontier topics" to convey a sense of the exciting new applications and rapid progress in this field. In particular, several chapters in this volume present important recently discovered computational innovations and research results. While it is quite possible that the field of computational economics will look very different 10 years from now than it does now, we designed this volume to provide a broad overview of the current state of the art, and a set of tools that will have a big impact on the way economics is done in the future.

Acknowledgements

We are indebted to many people who assisted in the production and editing of this *Handbook*. First of all we would like to thank the *Handbook* series editors, Kenneth Arrow and Michael Intriligator, for their constant support and encouragement. We

want to make special acknowledgement to John Geweke, Preston Miller, and Arthur Rolnick and the Federal Reserve Bank of Minneapolis for hosting the March 18–19, 1994 conference on Research and Training in Computational Economics, during which initial drafts of the chapters in this *Handbook* were presented and critiqued. We also want to thank the the Federal Reserve staff who helped organize the conference, including Carol Blunt, Jody Fahland, Correan Hanover, and Vicki Reupke. Finally we wish to acknowledge the diligence and promptness of the many “official” and unofficial referees who reviewed various versions of the chapters included in this *Handbook*. Their insightful comments greatly improved the overall quality of this volume (although of course we bear final responsibility for any of its defects). These referees included:

Sudhakar Achath, University of Texas, Sumru Altug, Virginia Polytechnic Institute, David Belsley, Boston College, Eric van Damme, Tilburg University, Hennie Daniels, Tilburg University, Paul Fisher, Bank of England, Manfred Gilli, University of Geneva, Vassilis Hajivassiliou, Yale University, Kenneth Judd, Hoover Institution, Michael Keane, University of Minnesota, Timothy Kehoe, University of Minnesota, Michel Keyzer, Center for World Food Studies, Blake LeBaron, University of Wisconsin, Martin Lettau, Tilburg University, Alan Manne, Stanford University, Ellen McGrattan, Federal Reserve Bank, Minneapolis, John Miller, Carnegie Mellon University, Reinhard Neck, University of Bielefeld, Soren Nielsen, University of Texas, Ariel Pakes, Yale University, Spassimir Paskov, Columbia University, Chris Phelan, MEDS, Northwestern University, Edward S. Prescott, University of Chicago, Martin Puterman, University of British Columbia, John Rowse, University of Calgary, Berc Rustem, Imperial College, University of London, Thomas Rutherford, University of Colorado, Herbert Scarf, Yale University, Sanjay Srivastava, Carnegie Mellon, Dale Stahl, University of Texas, Charles Tapiero, ESSEC, George Tauchen, Duke University, Joseph Traub, Columbia University, John Tsitsiklis, MIT, Marco Tucci, University of Siena, Kenneth Wallis, University of Warwick, Charles Whiteman, University of Iowa, Robert Wilson, Stanford University, Peter Zdrozny, GTE Labs.

H.M. Amman
University of Amsterdam

D.A. Kendrick
University of Texas

J. Rust
University of Wisconsin

References

- Heyman, D.P. and Sobel, M.J., eds (1990) *Handbooks in operations research and management science*. Amsterdam: Elsevier.
- Kendrick, D. et al. (1993) 'Research opportunities in computational economics', *Computational Economics*, 6:257–314.
- van Leeuwen, J., ed. (1990) *Handbook of theoretical computer science*. Amsterdam: Elsevier.